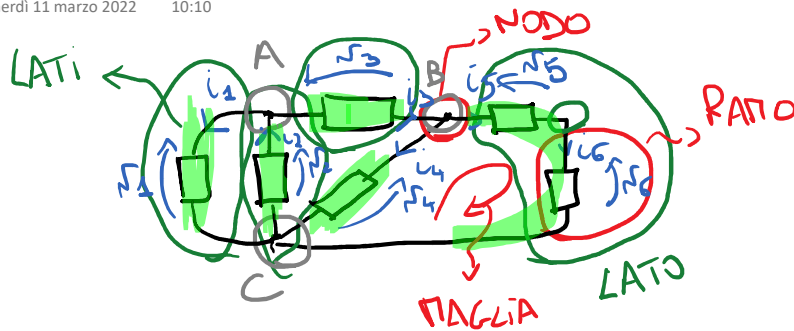


METODI DI ANALISI

venerdì 11 marzo 2022 10:10



LATO: ELEMENTO TOPOLOGICO
FORMATO DALLA SERIE DI
COMPONENTI CONNESSI A
DUE NODI FUNZIONALI

4 NODI
6 RATO
6 TAGLIA

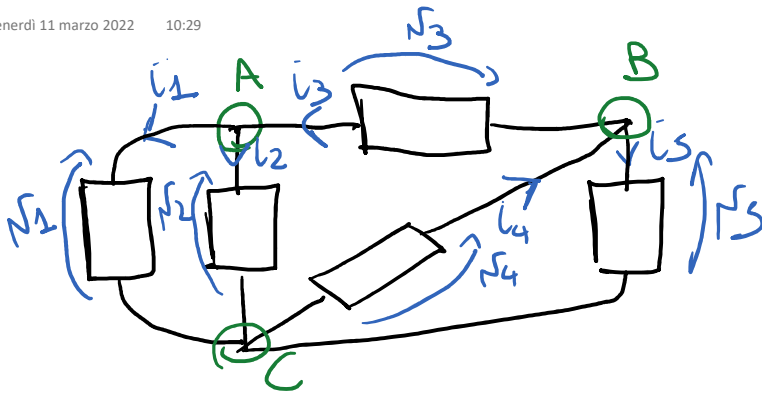
3 NODI
5 LATO
6 TAGLIA

RISOLVERE UN CIRCUITO SIGNIFICA TROVARE:

- TUTTE LE TENSIONI DI LATO
 - TUTTE LE CORRENTI DI LATO
- } **VARIABILI DI RETE**

$\Rightarrow 2 \cdot L$ ^{NO DI LATI} INCOGNITE $\rightarrow 2 \cdot L$ EQ. INDIPENDENTI

- EQ. COSTITUTIVE
- EQ. TOPOLOGICHE (LKT, LKC)

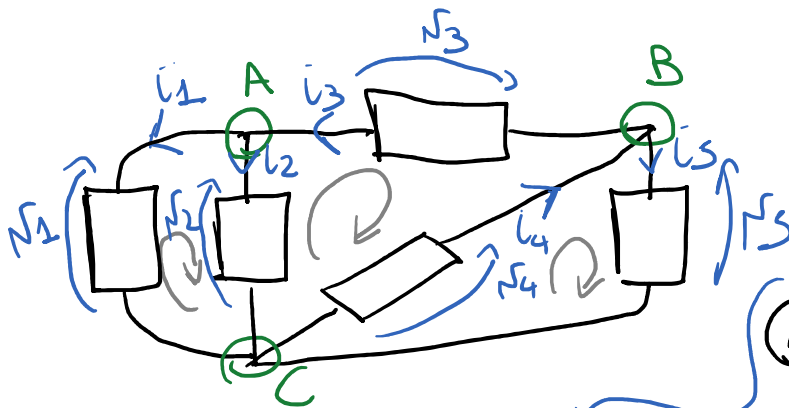


5 LATI $\left\{ \begin{array}{l} \rightarrow 5 \text{ TENSIONI} \\ \rightarrow 5 \text{ CORRENTI} \end{array} \right\}$
 3 NODI 10 INCOGN.

PER OGNI LATO/RAMO LA TENSIONE E LA CORRENTE
 DI LATO È VIOLATA DALL'EQ. COSTITUTIVA

L EQ. COSTITUTIVE

CI MANCANO ALTRE L EQ. INDIP. $\Rightarrow$ L_{KC}, L_{KT}



$N \text{ NODI} \rightarrow N \text{ LKC}$

$3 \text{ NODI} \rightarrow 3 \text{ LKC}$

(A) $i_3 - i_1 - i_2 = 0$

(B) $i_4 - i_3 - i_5 = 0$

(C) $i_1 + i_2 + i_5 - i_4 = 0$

$-i_4 - i_2 + i_4 - i_5 = 0$

$N \text{ NODI} \rightarrow N-1 \text{ LKC INDIPENDENTI}$

$L - (N-1) \text{ EQ. INDIP.} \Rightarrow L - N + 1 \text{ LKT INDIPENDENTI}$

SCEGLIAMO LE TAGLIE CHE NON SONO INTERSELTATE
DA RARI:

(1) $N_1 - N_2 = 0$

(2) $N_2 + N_3 - N_4 = 0$

(3) $N_4 - N_5 = 0$

METODO ALGEBRAICO DI TABLEAU

venerdì 11 marzo 2022

10:55



$L-N+1 \quad LKT$



RISOLVERE IL SISTEMA
DI $2L$ EQ. IN $2L$
INCONGITTE

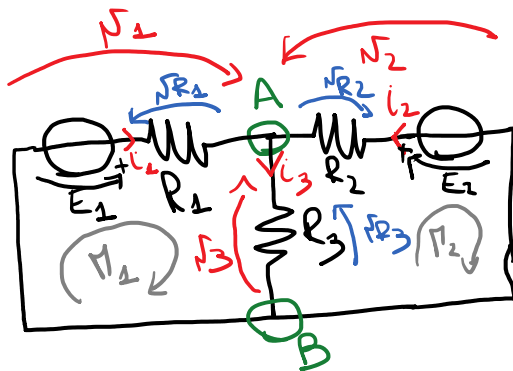


N, i

ESEMPIO

mercoledì 16 marzo 2022

16:13



2 NODI

3 LATI $\Rightarrow$ 2-3 INCOGNITE

EQ. COSTIT.

$$\text{LATO 1: } N_1 = E_1 - N_{R1} = E_1 - R_1 i_1$$

$$\text{LATO 2: } N_2 = E_2 - N_{R2} = E_2 - R_2 i_2$$

$$\text{LATO 3: } N_3 = N_{R3} = R_3 i_3$$

$$\text{LKC A: } i_1 + i_2 - i_3 = 0$$

$$\text{LKT } \pi 1: N_1 - N_3 = 0$$

$$\text{LKT } \pi 2: N_3 - N_2 = 0$$

$$\Rightarrow \begin{cases} i_1 + i_2 - i_3 = 0 \\ E_1 - R_1 i_1 - R_3 i_3 = 0 \\ R_3 i_3 - E_2 + R_2 i_2 = 0 \end{cases}$$

IPOTESI DI LINEARITÀ

$f(x_1 + x_2)$ è lineare:

ADDITIVITÀ: $f(x_1 + x_2) = f(x_1) + f(x_2)$

OMogeneità: $f(ax_1) = a f(x_1)$

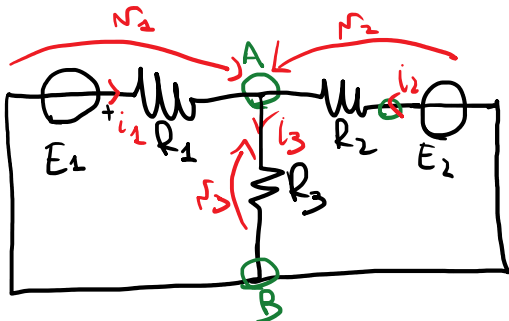
- SOVRAPPOSIZIONE DEGLI EFFETTI

IP: CIRCUITO LINEARE

Le variabili di rete (effetti) si possono ottenere come sovrapposizione delle risposte dovute alle singole cause.

Esempio

mercoledì 16 marzo 2022 16:52



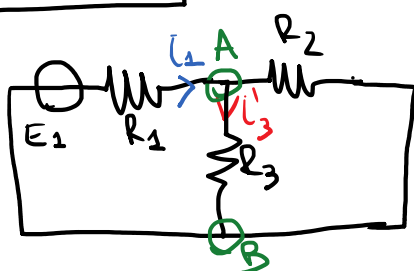
$$i_3 = ?$$

una causa alla volta $\Rightarrow$ PASSIVARE alcuni generatori:

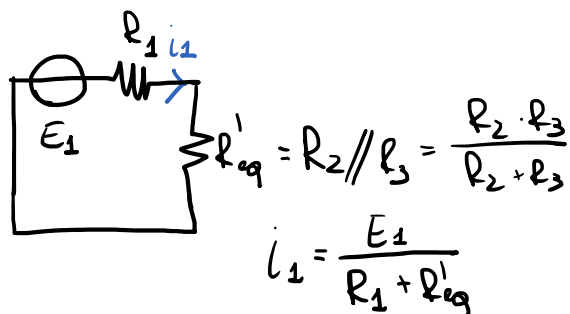
$$E_1 \rightarrow E_1 = \emptyset \rightarrow \text{CIRCUITO CHIUSO}$$

$$I_1 \rightarrow I_1 = \emptyset \rightarrow \text{CIRCUITO APERTO}$$

1) CAUSA E_1



$\equiv$

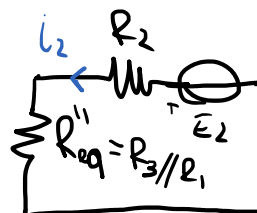


$$i_3' = i_1 \cdot \frac{R_2}{R_2 + R_3} = \frac{E_1}{R_1 + R'_{eq}} \cdot \frac{R_2}{R_2 + R_3}$$

2) CAUSA E_2



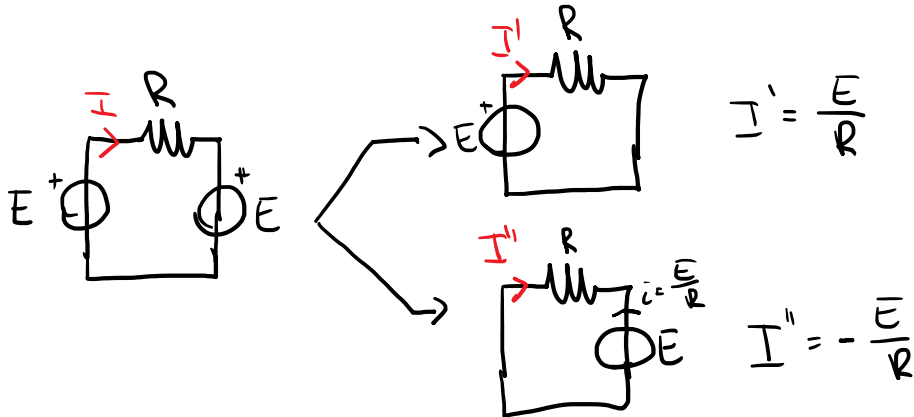
$\equiv$



$$i_2 = \frac{E_2}{R_2 + R''_{eq}}$$

$$i_3'' = i_2 \cdot \frac{R_1}{R_1 + R_3}$$

3) $\Rightarrow i_3 = i_3' + i_3''$



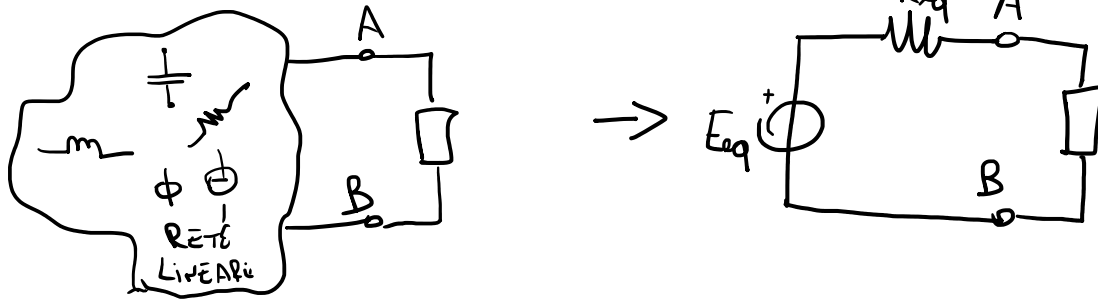
$$I = I' + I'' = \frac{E}{R} - \frac{E}{R} = 0$$

$$P = R I^2 = 0 \text{ W}$$

$$\cancel{P = P' + P''} = R I'^2 + R I''^2 = R \left(\frac{E}{R} \right)^2 + R \left(-\frac{E}{R} \right)^2 = R \frac{E^2}{R^2} + R \frac{E^2}{R^2} \neq 0$$

TEOREMA DI THEVENIN

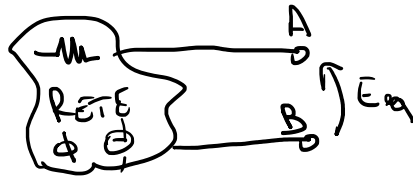
mercoledì 16 marzo 2022 17:16



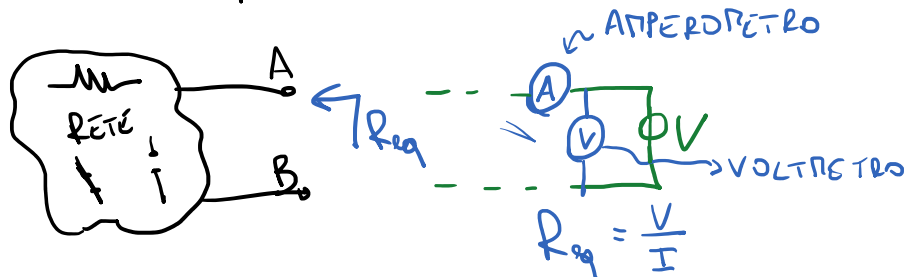
IP.: CIRCUITO LINEARE

IL CARICO NON È ACCORPIATO CON LA RETE DA SEMPLIFICARE

E_{eq} = è la tensione vista ai morsetti A e B a circuito aperto

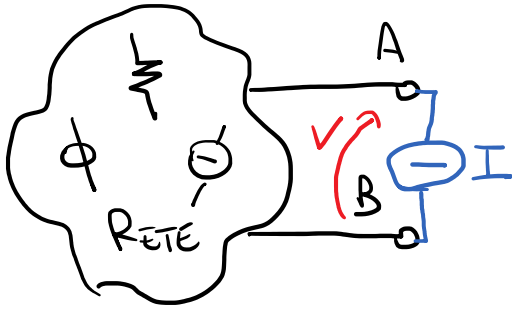


R_{eq} = è la resistenza equivalente vista da A e B partendo dai generatori indipendenti



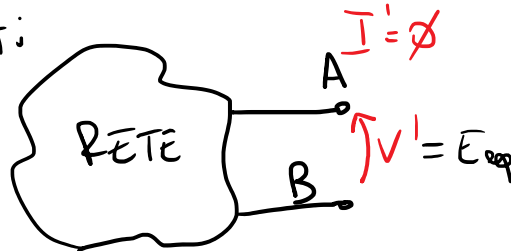
DIMOSTRAZ.

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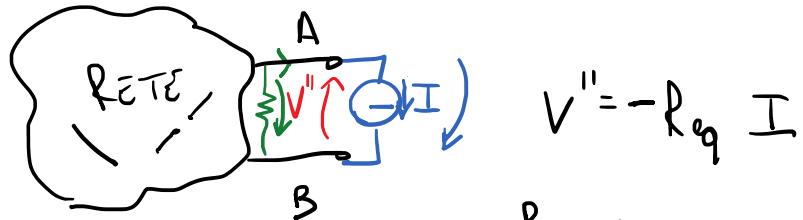


SOVRAPP. DEGLI EFFETTI:

- PASSIVANDO I:



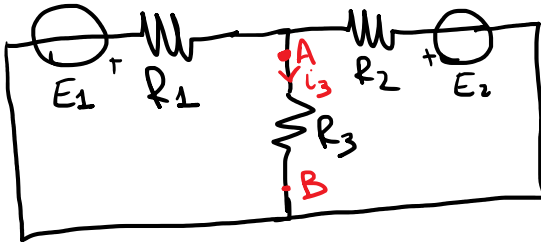
- PASSIVANDO LA RETE:



$$V = V' + V'' = E_{eq} - R_{eq} I \rightarrow$$

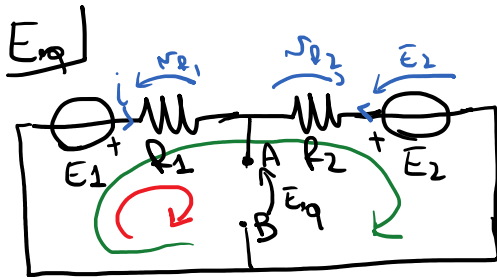
Esempio

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$$i_3 = ?$$

TEOREMA DI THEVENIN AI PUNTI A e B



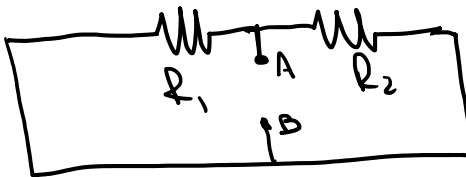
$$\text{LKT: } E_1 - iR_1 + iR_2 - E_2 = 0 \rightarrow E_1 - R_1 i - R_2 i - E_2 = 0$$

$$E_1 - E_2 = R_1 i + R_2 i \rightarrow E_1 - E_2 = (R_1 + R_2) i \rightarrow i = \frac{E_1 - E_2}{R_1 + R_2}$$

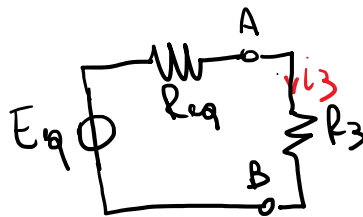
$$\text{LKT: } E_1 - R_1 i - E_{eq} = 0 \rightarrow E_{eq} = E_1 - R_1 \cdot \frac{E_1 - E_2}{R_1 + R_2} =$$

$$= \frac{\cancel{E_1} R_1 + E_1 R_2 - \cancel{R_1} E_1 + R_1 E_2}{R_1 + R_2} = \frac{E_1 R_2 + E_2 R_1}{R_1 + R_2}$$

R_{eq}



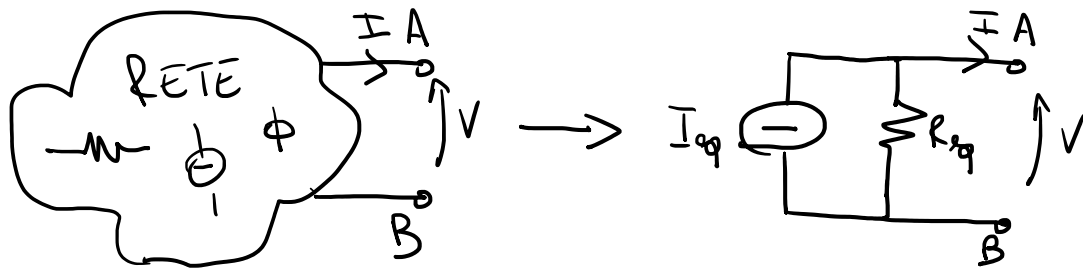
$$\equiv R_{eq} = R_1 // R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$$



$$i_3 = \frac{E_{eq}}{R_3 + R_{eq}} = \frac{E_1 R_2 + E_2 R_1}{R_1 + R_2} \cdot \frac{1}{R_3 + R_{eq}}$$

TEOREMA DI NORTON

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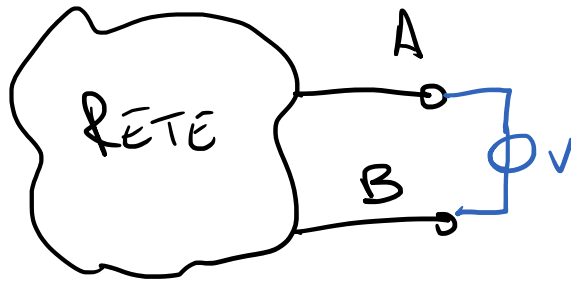
I_q : le stesse del Teorema di Thevenin

I_q = è la corrente fra A e B quando vi è un corto-circuito

R_q = è la resistenza vista ai morsetti A e B partendo dai generatori indipendenti

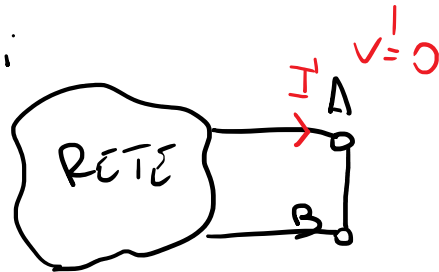
Dim.

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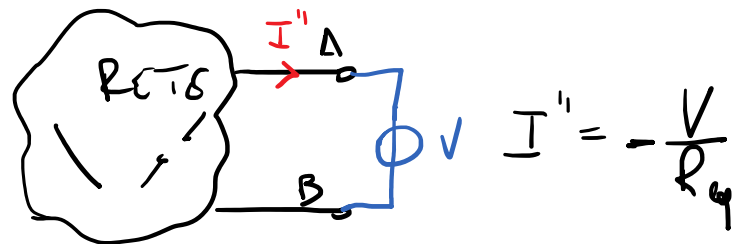


SOVRAPP. DEGLI EFFETTI:

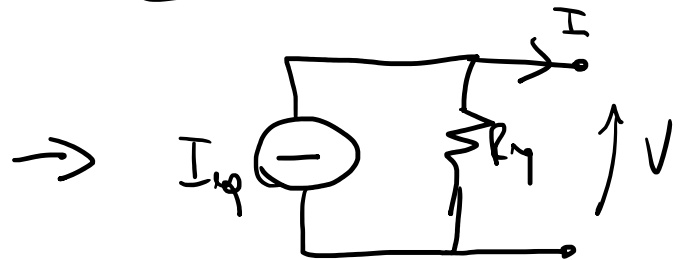
- PASSIVANDO V :



- PASSIVARE LA RETE:



$$\Rightarrow I = I_q - \frac{V}{R_q}$$

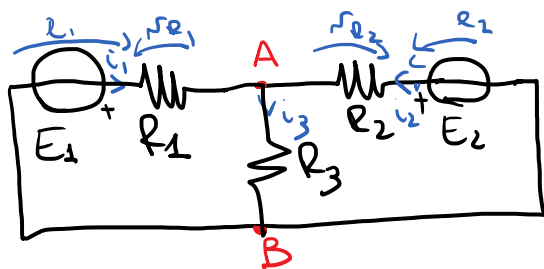


$$E_{eq} = R_{eq} I_q$$

$$R_{eq} = R_{eq}$$

Esempio

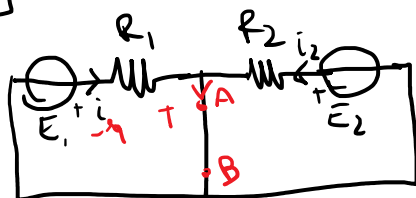
mercoledì 16 marzo 2022 18:21



$$i_3 = ?$$

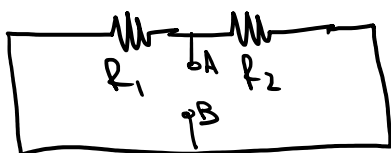
NORTON

I_{eq}

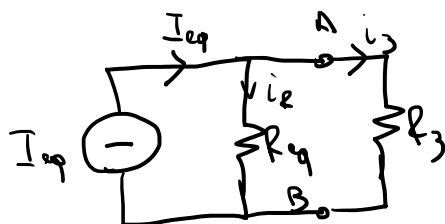


$$I_{eq} = i_1 + i_2 = \frac{E_1}{R_1} + \frac{E_2}{R_2}$$

R_{eq}



$$R_{eq} = R_1 // R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

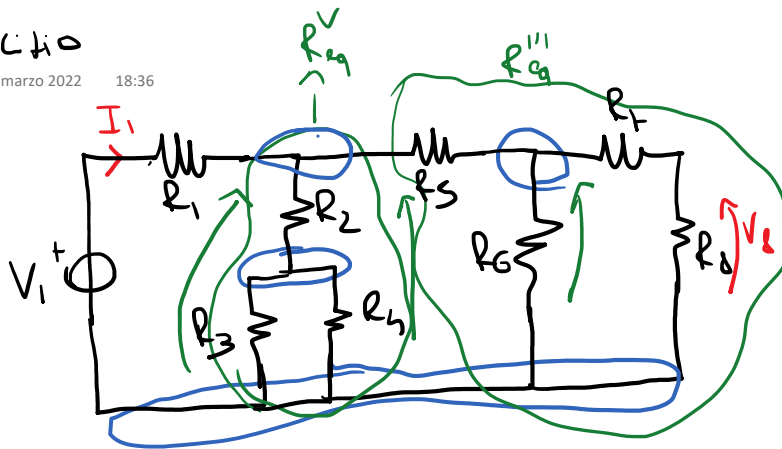


$$i_3 = I_{eq} \cdot \frac{R_{eq}}{R_{eq} + R_3} = \left(\frac{E_1}{R_1} + \frac{E_2}{R_2} \right) \cdot \frac{R_{eq}}{R_{eq} + R_3}$$

$$= \frac{E_1 R_2 + E_2 R_1}{R_1 R_2} \cdot \frac{R_1 R_2}{R_1 + R_2} \cdot \frac{1}{R_{eq} + R_3}$$

Esercizio

mercoledì 16 marzo 2022 18:36



$$V_1 = 15 \text{ V}$$

$$R_1 = 10 \Omega$$

$$R_2 = 25 \Omega$$

$$R_3 = 50 \Omega$$

$$R_4 = 20 \Omega$$

$$R_5 = 24 \Omega$$

$$R_6 = 20 \Omega$$

$$R_7 = 50 \Omega$$

$$R_8 = 30 \Omega$$

$$I_1 = ?$$

$$V_8 = ?$$

SEMPLIFICAZIONE

$$R_{eq}^I = R_7 + R_8 = 80 \Omega$$

$$R_{eq}^{II} = R_6 // R_{eq}^I = 16 \Omega$$

$$R_{eq}^{III} = R_5 + R_{eq}^{II} = 40 \Omega$$

$$R_{eq}^{VI} = R_{eq}^{III} // R_4 = 19,8 \Omega$$

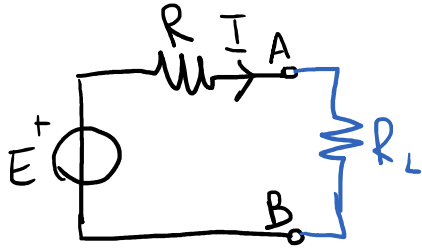
$$R_{eq}^{IV} = R_3 // R_4 \approx 14,3 \Omega$$

$$R_{eq}^V = R_2 + R_{eq}^{IV} = 39,3 \Omega$$

$$\Rightarrow I_1 = \frac{V_1}{R_1 + R_{eq}^{VI}} = \frac{15}{29,8} \approx 0,5 \text{ A}$$

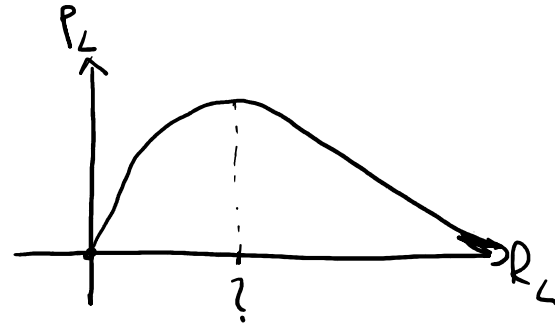
TEO. DEL MAX TRASFERIMENTO DI POTENZA

venerdì 18 marzo 2022 09:22



Quale valore di R_L per avere P_L MAX?

$$P_L = R_L \cdot I^2 = R_L \left(\frac{E}{R + R_L} \right)^2$$



$$\frac{\partial P_L}{\partial R_L} = 0$$

$$\frac{\partial P_L}{\partial R_L} = E^2 \frac{\partial}{\partial R_L} \left[\frac{R_L}{(R + R_L)^2} \right] = E^2 \left[\frac{(R + R_L)^2 - 2R_L(R + R_L)}{(R + R_L)^4} \right] = 0$$

$$R^2 + \cancel{R_L} + \cancel{2R_L} - \cancel{2R_L} - 2R_L^2 = 0 \rightarrow \boxed{R = R_L}$$

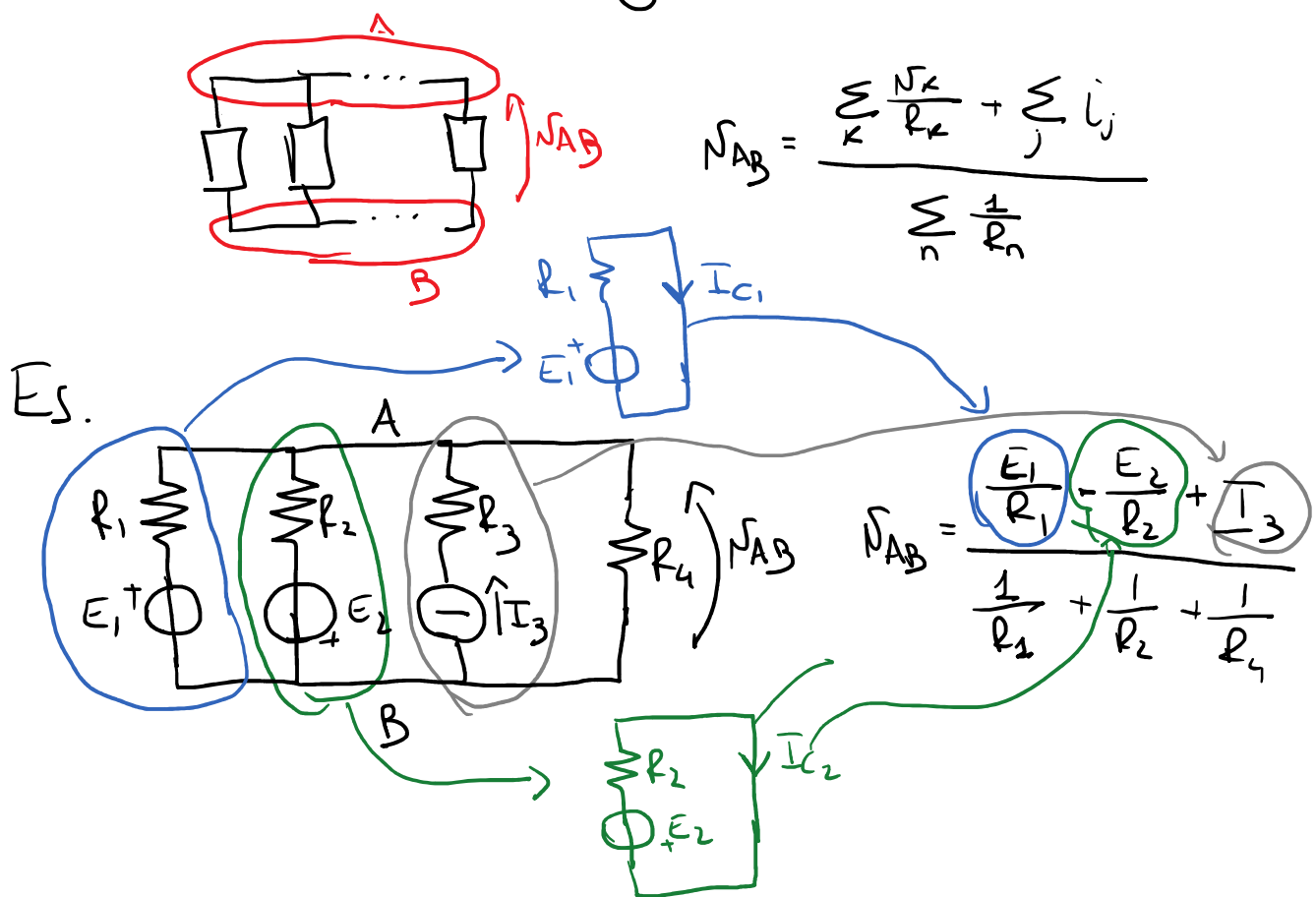
$$P_{L, \text{MAX}} = R \frac{E^2}{(R + R)^2} = R \frac{E^2}{4R^2} = \frac{E^2}{4R}$$

TEOREMA DI MILLMAN

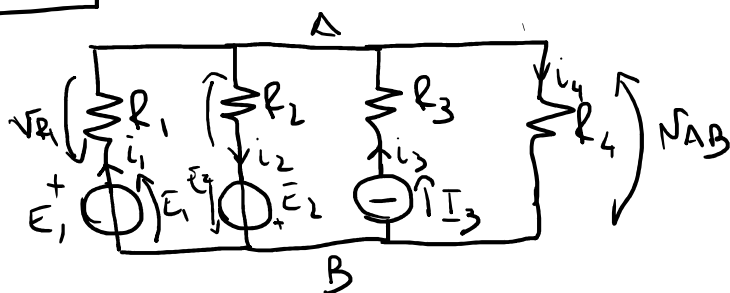
venerdì 18 marzo 2022 09:33

I.P.: LINEARITÀ

Dato una rete con 2 o più rami in parallelo, la Tensione ai capi della rete è pari al rapporto delle correnti di corto-circuito di ogni singolo ramo e la somma delle conduttanze di ogni ramo



Dim.



$$\textcircled{1} \quad N_{AB} = E_1 - R_1 i_1 \rightarrow i_1 = \frac{E_1 - N_{AB}}{R_1}$$

$$\textcircled{2} \quad N_{AB} = -E_2 + R_2 i_2 \rightarrow i_2 = \frac{E_2 + N_{AB}}{R_2}$$

$$\textcircled{3} \quad i_3 = I_3$$

$$\textcircled{4} \quad i_4 = \frac{N_{AB}}{R_4}$$

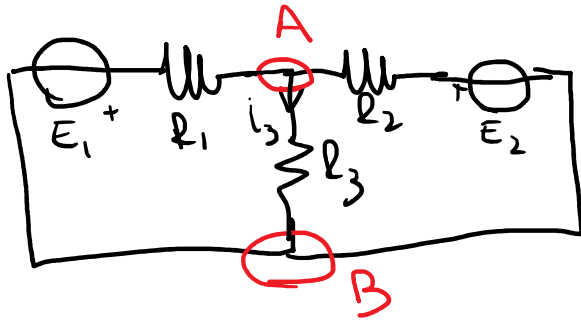
$$\text{LKC: } i_1 - i_2 + i_3 - i_4 = 0 \Rightarrow \frac{E_1 - N_{AB}}{R_1} - \frac{E_2 + N_{AB}}{R_2} + I_3 - \frac{N_{AB}}{R_4} = 0$$

$$\frac{E_1}{R_1} - \frac{N_{AB}}{R_1} - \frac{E_2}{R_2} - \frac{N_{AB}}{R_2} + I_3 - \frac{N_{AB}}{R_4} = 0$$

$$N_{AB} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4} \right) = \frac{E_1}{R_1} - \frac{E_2}{R_2} + I_3 \rightarrow N_{AB} = \frac{\frac{E_1}{R_1} - \frac{E_2}{R_2} + I_3}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_4}}$$

E_s .

venerdì 18 marzo 2022 09:58



$$i_3 = ?$$

$$N_{AB} = \frac{\frac{E_1}{R_1} + \frac{E_2}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}} \rightarrow i_3 = \frac{N_{AB}}{R_3}$$